Ashi Poorey

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EDUCATION

Bachelor of Science - Astrophysics *University of Illinois Urbana Champaign*
Minors - Data Science and Computational Science & Engineering *May 2027*

RESEARCH

Space Astronomy Summer Program intern - STScI *Jun 2026 - Aug 2026*
Undergraduate Researcher [REU]

- Developed machine learning models to identify rare galaxies exhibiting broad Mg II emission in Sloan Digital Sky Survey spectra, targeting signatures of fading supermassive black holes
- Analyzed large spectroscopic datasets (~1.4 million sources) to isolate and validate a small sample of candidate objects, enabling study of black hole accretion duty cycles
- Designed scalable methods for deployment on future surveys (SDSS extensions, DESI, etc.) to discover and characterize transient active galactic nuclei in time-domain astronomy

UIUC Team Gaia - Prof Jiayin Dong *Aug 2025 - present*
Undergraduate Researcher

- Developed an exoplanet detection pipeline using ESA Gaia mission data, reconstructing existing Bayesian inference code from PyStan to PyTensor + PyMC.
- Applied statistical methods to time-series astrometric data for exoplanet detection, with Bayesian statistics and probabilistic modeling
- Led research paper discussions and participated in hackathon-style coding sessions, strengthening skills in Python programming, scientific problem-solving, and teamwork.

UIS Stellar Evolution Research - Prof John Martin *Feb 2026 - present*
Undergraduate Researcher

- Research on Warm Hypergiants, investigating their variability and evolutionary behavior
- Analyzed time-series photometric features using Lomb–Scargle periodograms to identify periodic and quasi-periodic variability in M31 and M33 massive stars
- Interpreted variability trends in the context of stellar pulsations, mass loss, and late-stage massive star evolution; contributing to ongoing analysis for presentation and publication

CONFERENCES, TALKS & PRESENTATIONS

1. **APS CU*IP 2026** - Gaia Pipeline for Inferring Planetary Companions using RUWE *[poster]*
2. **CAURS 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 *[poster]*
3. **UIUC URS 2026 -** Gaia Pipeline for Inferring Planetary Companions using RUWE *[poster]*
4. **UIUC Astrofest 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 *[poster]*

PROGRAMS & TRAINING

STScI JWST Summer School – *High-Redshift Transients and Time Domain Astronomy* *Aug 2025*

- Selected to participate in a program combining lectures on high-z transient science with hands-on training in JWST observation planning, calibration pipelines, and data analysis.
- Gained practical experience in creating JWST observing programs and processing early-universe astrophysical data using advanced infrared techniques.

AWARDS

1. **Distinguished Undergraduate Researcher -** University of Illinois Urbana Champaign
2. **Celia Mathews Elliot Prize for Writing in Science and Engineering** - Department of Physics UIUC

OUTREACH, VOLUNTEERING & TEACHING

The Jeffries Academic Center - *Math & Physics Tutor* *Jan 2025 - May 2025*
Provided weekly academic support in Physics and Math to over five students

UIUC College of Liberal Arts & Sciences - *Student Ambassador* *Sep 2025 - present*
Representing the college at outreach and media events, and assisting prospective students with academics, campus life, and opportunities in LAS

UIUC LAS Lift Off 2025 - *Department of Astronomy booth* *Aug 2024*
Represented the Astronomy department, welcoming new students and answering questions about courses, research opportunities, and campus resources

Illinois Space Society - *Peer Mentor* *Aug 2024 - Dec 2024*
Mentored a freshman student by providing guidance on academics, opportunities, and career advice

PROJECTS

Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics **|** *Publication*
Co-authored research on diversity and retention in astrophysics, synthesizing literature and community insights into actionable recommendations

Radio Astronomy Capstone Project: S255IR NIRS3
Processed ALMA data using CASA and CARTA to produce continuum and C34S molecular line results, generating publication-quality figures and deriving astrophysical interpretations consistent with observational studies

TECHNICAL SKILLS

Languages: Python, C++, SQL, MATLAB, Linux, Arduino
Libraries: AstroPy, NumPy, SciPy, Pandas, Seaborn, Statsmodel, PyStan, PyTensor, Scikit, PyTorch